FINA6229
Machine Learning in Finance

Project 2
Due Date: 12:00 PM, 26 March 2025
Individual Stock Return Predictability and Factor Investing

Instruction: You are encouraged to work in groups (but not across groups). However, you should independently write up and submit your own assignments through the CUHK Blackboard under the folder: *FINA6229FA>>Course Contents>>Lecture 2>>Project 2*. Demo code (as an image file) is provided under the same folder. In addition to answering the questions below, you are asked to **add some comments right after the "#" whenever it appears in the demo code**, to explain the purpose of the corresponding code.

As an example,

```
#
monthly = pd.read_csv('monthly2020.csv', index_col=0, header=None)
T = monthly.shape[0]
```

you are supposed to submit the code with some comments as follows:

```
# import monthly data using the pandas module
monthly = pd.read_csv('monthly2020.csv', index_col=0, header=None)
T = monthly.shape[0]
```

Introduction

The second project is aimed at helping you better understand the process of constructing investment "factors" from the returns of individual stocks. Essentially you are required to replicate some asset pricing anomalies and implement various hedge-fund trading strategies. You are supposed to process the sample data to form portfolios consisting of stocks with similar characteristics, and then evaluate the performance of various long-short portfolios (monthly rebalanced). The explicit goal is to replicate three popular factors in the academic literature.

Note that in practice, there is no fully agreed definition for a given factor. For example, many quant hedge funds claim to apply momentum investing, but they may define momentum in different ways, leading to diverse trading decisions. Different venders such as S&P Capital IQ (Alpha Factor Library), Factset (Alpha Testing), Bloomberg (FTST), or MSCI (Barra) may provide somewhat different numbers for the same factor even if they use the same definition, due to some subtle differences listed below:

- **Data processing**

  How they deal with missing value, outliers, point-in-time, portfolio weighting, holding periods, turnover, and survivorship bias.
- **Sample coverage**

  The universe of stocks that are used to construct the factors may differ, such as different indices (S&P500, Russell 3000, MSCI) and different exchanges (NYSE, NASDAQ, AMEX, OTC).
- **Time-period selection**

  For example, some use the latest quarterly EPS to compute PE ratio for convenience, while others use a 12-month moving-average EPS to smooth the data.

To facilitate the construction of the factors, a well-organized dataset including all potential variables you need is provided. In the real world, it is rarely the case that you will be asked to begin the work by collecting firm-level data. Instead, most quantitative funds either directly buy organized data from data vendors or use a third-party platform. However, it is still helpful for you to understand how the data is processed and what issues should be paid attention during the backtesting.

I hope this project will also help improve your skills to deal with large dataset in Python. The best way to learn a programming language is to use it. Internet is the best tool when you confront any programming issues. When you get stuck or are unsure about how to use specific functions, try Python Help and Documentation.

Data Description

The sources of the dataset we will use come from a commonly used database *Compustat-Capital IQ*, which provides firm-level fundamental and security data. While fundamental data is organized annually and quarterly, the security trading data is collected on either monthly or daily basis.

It is worth mentioning that different data vendors often use their own unique identifier to track a company or a security. You may be more familiar with *ticker* used by stock exchanges. For backtesting, since we need to consider issues such as delisting, M&A, and cross listing, the best way is to use the unique identifier provided by those data vendors. For example, Compustat uses *gvkey* as the unique identifier for each security/firm. We will use *gvkey* from Compustat as the unique identifier for this project.

The sample coverage includes all companies that have been a member of S&P 500. You will see there are actually more than 500 stocks in the dataset due to changes in the constituents of

S&P 500 over time. For your convenience, the index indicator is provided as a separate variable (***sp500***) in the dataset (member=1, nonmember=0 in a given month).

Last but not least, the historical returns data are not perfect. For example, missing values are quite common. There could even be reporting errors (check for extreme outliers in the data).

Replication Process

1. Definition of the factors

Below are three factors (**variable names are bolded in the parentheses**) selected from *Capital IQ Alpha factor Library* for you to replicate. You are required to replicate all of them. The time period of the replication is the same as the *Capital IQ* benchmark: **198701~202212.**

1) Size: Log Market Cap

$$LogMCap_{i,t} = Log\left(CSHOM_{i,t} \times CloseM_{i,t}\right)$$

CloseM: Adjusted Closing Price (**prccm**)
CSHOM: Adjusted Monthly Common Shares Outstanding (**cshom**)
Benchmark in Capital IQ: **LogMktCap**

2) Valuation: Book to Price

$$BP_{i,t} = \frac{CEQQ_{i,t}}{CSHOQ_{i,t} \times CloseM_{i,t}}$$

CloseM : Adjusted Closing Price (**prccm**)
CEQQ: Quarterly Common Equity in millions (**ceqq**)
CSHOQ: Adjusted Quarterly Common Shares Outstanding in millions (**cshoq**)
Benchmark in Capital IQ: **BP**

3) Price Momentum: 1M price High-1M price Low

$$HL1M_{i,t} = \frac{HighM_{i,t} - CloseM_{i,t}}{CloseM_{i,t} - LowM_{i,t}}$$

CloseM : Adjusted Closing Price (**prccm**)
HighM : Adjusted Monthly High (**prchm**)
LowM : Adjusted Monthly Low (**prclm**)
Benchmark in Capital IQ: **HL1M**

2. Portfolio sorts and construction of factors

1) Downloading Data

The dataset can be downloaded from the CUHK Blackboard. The dataset contains three tables: **Benchmark, Data, and S&P 500 Index:**

- **Benchmark2022.csv**: is the benchmark of the replication exercise. Your output should be similar to the factors contained in this dataset (e.g., the factor you constructed should have high correlations with the corresponding factors provided by Capital IQ).
- **Data2022.csv:** include all variables (e.g., individual stock returns and stock characteristics) you need to replicate the three investment factors specified above.
- **SP500_2022.csv:** includes the total monthly returns of S&P 500 Index and one-month T-Bill rate as risk-free rate.

Store the dataset files under your local working directory. Note: we will use the sample period **between 197001 and 202212** for the portfolio sorting tests.

2) Process data into an organized form

There is no standard way to process data. You can use any structure to organize data.

3) Sorting

Stocks are sorted into **quintiles** (i.e., 5 bins) based on the values of various stock characteristics (see the definition of the three factors above).

4) Evaluate the portfolio performance

Portfolios are rebalanced once a month. All portfolios are **equal weighted**. To calculate the returns of your portfolios, you only need to take the average returns of the stocks in each of the quintile portfolio. For simplicity, the monthly total returns of stocks (including both capital gain and dividends, adjusted for corporate events such as stock splits) are provided in the dataset (**trt1m**). The total returns are displayed in percentage.

5) Compute QSpread

For consistency, the quantile spread is defined as the difference of the returns between the top and bottom portfolio in each month. $\boldsymbol{QSpread = Top - Bottom}$

Questions

Q1: Choose "Book to price (BP)" as an example to answer this question.

- Plot the times series of the number of firms in your sample (Note: Due to the quality of the data, the number of firms used in the sample may not always be 500 but should be around 500.)
- Plot the time series of the BP QSpread you obtained and compare it to the corresponding factor from Capital IQ.
- Plot the returns of the long-leg and short-leg of the strategy separately.
- Plot the **cumulative** returns for BP QSpread, long-leg, short-leg, and S&P 500 index over the sample period.
- Please comment shortly on the performance of the factor BP and compare it to S&P 500. Are their performances comparable? Why or why not? Which contributes to the factor performance more, the long-leg or the short-leg? Can you find any pattern in when (in which time periods) the factor return tends to be poor? (e.g., recession vs. booming periods)

Q2: Report the correlations of your factors and the corresponding benchmarks from Capital IQ. If your code is correct, **all correlations should be about 0.9 or greater**. If the correlation you calculated is negative, just take the absolute value (i.e., switch the long side vs. short side of your factor). **Note that for this question, use the sample period of Capital IQ: 198701 to 202212**

Table: Correlations between the replication and the benchmark, 198701~202212		
Factor	**Description**	**Correlation**
Size	Log Market Cap	
BP	Book to Price	
HL1M	1M Price High- 1M Price Low	

Q3: Calculate the following statistics for all factors you have constructed: **mean, standard deviation, Sharpe ratio, skewness, kurtosis, maximum, minimum, autocorrelation, and correlation matrix (Note that for this question, use the whole sample period 197001 to 202212).**

Please shortly comment on their performance based on those statistics. What are the advantages and disadvantages of those factors? How do they perform compared to S&P 500 index? If you are asked to recommend one investment factor to your portfolio manager, which one will you choose?

Factor	**Size**	**BP**	**HL1M**	**S&P 500**
Mean (%)				
STD (%)				
Sharpe Ratio				
Skewness				
Kurtosis				
Max (%)				
Min (%)				
ACF 1				
ACF 12				
ACF 24				

Table: Correlation Matrix, 197001~202212				
	Size	BP	HL1M	S&P 500
Size				
BP				
HL1M				
S&P 500				

Q4: Conduct a hypothesis test for each factor you constructed. Using the time-series of the factor returns (QSpread) and a t-test to show whether the mean of each factor is significantly different from zero. Please highlight those factors with significant QSpread. Are they consistent with your expectation? **(Note that for this question, use the whole sample period: 197001 to 202212)**

Table: testing the significance of factor mean return, 197001~202212				
Factor	**Name**	**Average Spread (%)**	***t* statistics**	***p*-value**
Size	Log Market Cap			
BP	Book to Price			
HL1M	1M Price High- 1M Price Low			
S&P 500	S&P 500 Index			

Lastly, please submit this assignment electronically through the CUHK Blackboard as **a single Jupyter Notebook (*.ipynb) with all the outputs and the answers** to the questions above and the code outputs. **Please name the submitted file with project number, your full name, student ID, and group number as**:

ProjectNumber_StudentName_StudentID_GroupNumber.ipynb,

for example: *Project1_GangLi_12345678_Group1.ipynb*.